Skills Summary

Binomial Coefficients in Probability and Expansion

Use this reference while completing your worksheet or when studying.

Skill 1 — Binomial coefficients and Pascal's rows

Formula: $\binom{n}{k} = \dfrac{n!}{k!\,(n-k)!}$ — the number of ways to choose k things from n. **Addition rule:** $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$, which is how each Pascal entry is the sum of the two above it. Also $\binom{n}{k} = \binom{n}{n-k}$, so every row is symmetric.

Example: Evaluate $\binom{7}{3}$.

Step 1. This is a straight count of choices, so apply the factorial formula and cancel before multiplying.

Step 2. $\dfrac{7!}{3!\,4!} = \dfrac{7\cdot 6\cdot 5}{3\cdot 2\cdot 1}$.

$\Rightarrow \quad \binom{7}{3} = \mathbf{35}$

Try it: Evaluate $\binom{8}{2}$.

check: **28**

Skill 2 — Expanding a binomial power

Binomial theorem:

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$$

The exponents on a count down while those on b count up, and every term's exponents add to n. The coefficients are row n of Pascal's triangle.

Example: Expand $(x+1)^3$.

Step 1. A whole power of a binomial with a small exponent — read the coefficients straight off Pascal's row 3 rather than multiplying out.

Step 2. Row 3 is $1, 3, 3, 1$, so the terms are x^3, $3x^2$, $3x$, 1.

$\Rightarrow \quad (x+1)^3 = \boldsymbol{x^3 + 3x^2 + 3x + 1}$

Try it: Expand $(x-2)^3$.

check: $\boldsymbol{x^3 - 6x^2 + 12x - 8}$

Skill 3 — Picking out one term

Rule: the term of $(a+b)^n$ containing b^k is $\binom{n}{k}a^{n-k}b^k$. Choose k from the power you want, then let $n-k$ fill the other exponent — never expand the whole thing when one term is asked for.

Example: Find the coefficient of x^2 in $(x+2)^4$.
Step 1. Only one term carries x^2, so match exponents first: x^2 needs the 2 used twice, giving $k=2$.
Step 2. The term is $\binom{4}{2}x^2 \cdot 2^2 = 6 \cdot x^2 \cdot 4$.
$\Rightarrow$ coefficient $=$ **24**

Try it: Find the coefficient of x^3 in $(x+1)^6$.

check: **20**

Skill 4 — Binomial probability

Rule: for n independent trials each succeeding with probability p,

$$P(\text{exactly } k \text{ successes}) = \binom{n}{k}p^k(1-p)^{n-k}.$$

The power gives the chance of one particular ordering; the coefficient counts how many orderings there are. For "at least k", add the separate terms.

Example: A fair coin is flipped 4 times (16 equally likely sequences, 6 of them with exactly two heads). Find P(exactly two heads).
Step 1. All sequences are equally likely, so the probability is a plain count of favourable sequences over all sequences.
Step 2. $P = \frac{6}{16}$, which reduces by 2.
$\Rightarrow$ $P = \boldsymbol{\frac{3}{8}}$

Try it: Same 4 flips; 4 of the 16 sequences have exactly one head. Find that probability.

check: $P = \boldsymbol{\frac{1}{4}}$